SAUMIT PAUL

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SUMMARY

Electrical Engineering M.S. candidate (**May 2027 Grad**) with 3 years of experience building, validating, deploying, and monitoring production machine learning systems. Experienced in computer vision, deep learning, model evaluation, data pipelines, model optimization, and LLM-based agent systems.

PROFESSIONAL EXPERIENCE

Data Scientist, DeepTek.ai 06/2022 - 07/2025

- Developed and validated PyTorch and TensorFlow/Keras models for chest X-ray image classification and semantic segmentation, including pleural effusion detection using CNN backbones (ResNet, DenseNet, U-Net, FPN) achieving over 90% AUROC on held-out test data.
- Owned end-to-end computer vision lifecycle from dataset curation, image preprocessing and augmentation (OpenCV, normalization, histogram equalization, random crop/flip/rotation), training, validation, ONNX export, deployment, production monitoring, and failure analysis.
- Built automated computer vision evaluation and visual debugging platform with batch inference, MLflow tracking, computer vision metrics (AUROC, Dice coefficient, IoU) and visual error analysis, reducing evaluation from several hours to under 5 min.
- Maintained and validated a MongoDB-backed annotation and metadata system covering more than 1.8 million imaging studies, resolving schema and data-quality issues across heterogeneous clinical datasets.
- Applied INT8 quantization-aware training with TensorFlow Model Optimization and measured the resulting accuracy and deployment-efficiency tradeoffs for a pleural effusion model.
- Collaborated with engineering, clinical, regulatory, product, and customer-facing teams to investigate model failures, explain behavior, support demonstrations, and translate requirements into model changes.

PROJECTS

Security-Utility Tradeoffs in Coding Agents (GitHub) 09/2026 - Present

- Investigated security-utility tradeoffs in LLM coding agents by implementing task-aware command filtering using Jev alongside deterministic regex-based guardrails.
- Co-developed a Docker-isolated evaluation framework integrating mini-swe-agent with CodeIPI, SWEBench-Verified and IssueTrojanBench, supporting 144 prompt-injection scenarios and Langfuse tracing of agent actions, guardrail decisions, and benchmark scores.

LLM-Based Cache Replacement Policy Optimization 08/2025 - 12/2025

- Built an autonomous LLM harness to generate, compile, benchmark, and refine C++ cache replacement policies in ChampSim using SPEC CPU 2006 IPC feedback, reaching an average IPC of 0.414.
- Implemented graph-structured retrieval over prior policies and early rejection using representative workloads, reducing evaluation time from roughly 40–50 minutes to 30 minutes.

SetScout: Agentic Dataset Retrieval and Ranking (GitHub) 08/2026 - Present

- Developed a LangGraph agent that converts natural-language dataset requirements into structured searches across Hugging Face and Kaggle, retrieves documentation, and ranks candidates against user requirements.
- Implemented asynchronous multi-source retrieval, structured Pydantic outputs, configurable candidate limits, and error handling for reliable agent execution.

EDUCATION

North Carolina State University 08/2025 - **05/2027**
Master of Science, Electrical Engineering GPA: 4.0
Relevant Coursework: Generative AI for Systems, Trustworthy AI for Software Engineering, Neural Networks

Manipal Institute of Technology 07/2018 - 08/2022
Bachelor of Technology, Electrical and Electronics Engineering WES GPA: 3.8

SELECTED PUBLICATIONS

- A Comprehensive Evaluation of DeepTek CXR Analyzer in Detecting and Localising Suspicious Findings in Chest X-rays (ECR 2024)
- Artificial Intelligence as a Proficient Tool in Detecting Pulmonary Tuberculosis in Massive Population Screening Programmes (Journal of Rural Medicine)

SKILLS

Programming	Python, C++, SQL, Bash
Machine Learning	PyTorch, TensorFlow/Keras, Hugging Face Transformers, Scikit-Learn
LLM / Agents	LangGraph, LangChain, RAG, Tool Calling, Pydantic, OpenRouter
Evaluation / MLOps	MLflow, Langfuse, ONNX, DVC, Docker, Git, Linux
Data	MongoDB, Pandas, NumPy, Azure Data Factory